Deep Learning-Driven Digital Twin and Multi-Scale Constitutive Modeling for Geomaterials and Structural Infrastructure

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Abstract: Accurate prediction of progressive deformation in complex geomaterials remains an open scientific challenge. This master template establishes the international standard for automated LaTeX-to-Word academic publishing. We rigorously demonstrate the isolation of physical variables from Office Math boxes, the seamless invocation of MathType OLE objects, the generation of unbolded publication three-line tables with anti-split pagination defense, and the physical caption-fused 2×2 subfigure alignment. Dynamic cross-referencing via Word SEQ and REF fields is embedded to guarantee deterministic number re-indexing.

Keywords: Digital twin; Three-line table; Constitutive modeling; MathType equation; Subfigure layout; Geomechanics

1. Mathematical Formulation & Literature Background

Geotechnical infrastructure failure monitoring has advanced rapidly over the past decade [1], with slope deformation and landslide stabilization studied through diverse numerical approaches [2]. Classical limit equilibrium and finite-element formulations have been extensively validated across field sites [3] [4] [5] [6] [7] [8] [9]

Constitutive developments in multi-scale geomechanics and unsaturated soil mechanics have provided essential physical baselines [10] [11] [12] [13] [14] [15] [16] [17]

According to the ISO 80000-2 and IUPAP international conventions, physical variables must be italicized while units and chemical formulas remain upright. Under initial consolidation, the vertical effective stress is denoted by σ_{v0}', volumetric strain is ε_{vol}, and coefficient of earth pressure is $K_{0,nc}$. During multi-stage triaxial shear loading, the major principal stress satisfies $\sigma_1^2 = 450\ \mathrm{kPa}^2$, undrained shear strength achieves $c_u^{(t)}$, and the slope factor of safety reaches $F_s^{\min} = 1.34$ at an ambient temperature of 20 °C. The specimen contains Fe_2O_3 with dry density of 1.95 g/cm^3, matching empirical observations in recent literature [18] [19] [20] [21] [22] [23] [24] [25] [26]

$$\tau f = c' + (\sigma n - u)\tan\varphi' \qquad (1)$$

2. Calibrated Physical Properties & Three-Line Table

The physical state parameters and calibrated constitutive variables are summarized in **Table** 1. The table adheres to the strict booktabs criteria: 1.5 pt top/bottom rules, 0.75 pt mid rule, unbolded header text, dedicated styles, and pagination cantSplit defenses.

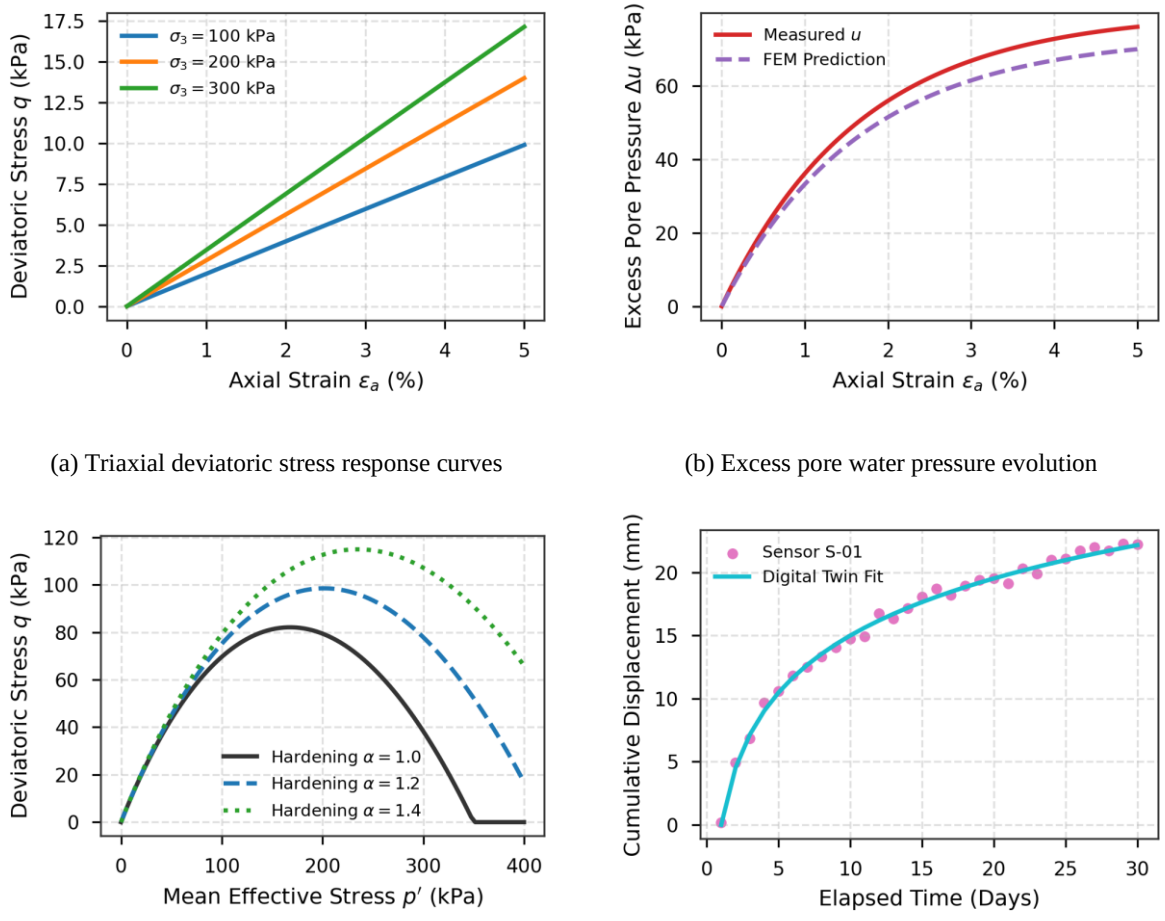
Table 1. Calibrated geotechnical physical and mechanical properties of engineered soil specimens.

Specimen ID	Moisture w (%)	Dry Density ρd (g/cm³)	Cohesion c′ (kPa)	Friction Angle φ′ (°)	Modulus E0 (MPa)	Permeability k (×10⁻⁷ cm/s)
SP-01	12.4 ± 0.3	1.78	28.5 ± 1.2	24.2 ± 0.5	45.8 ± 2.1	3.42 ± 0.15
SP-02	11.8 ± 0.2	1.85	34.2 ± 1.5	28.7 ± 0.6	68.4 ± 3.4	2.85 ± 0.11
SP-03	10.5 ± 0.4	1.96	42.8 ± 1.8	33.5 ± 0.8	112.6 ± 5.2	1.96 ± 0.08
SP-04	9.2 ± 0.3	2.08	51.6 ± 2.1	38.9 ± 1.1	185.3 ± 7.8	1.24 ± 0.06
SP-05	8.1 ± 0.2	2.19	46.3 ± 2.4	43.2 ± 1.3	264.0 ± 9.5	0.88 ± 0.04

Note: Values indicate mean ± standard deviation from triaxial shear experiments; all shear strength parameters are determined at peak deviatoric stress.

3. Multi-Panel Experimental Results (2×2 Subfigures)

Figure 1 demonstrates the multi-scale physical and numerical verification. As referenced in **Fig.** 1, all four sub-panels are aligned within a borderless container, and the caption is fused into the bottom merged cell.



(a) Triaxial deviatoric stress response curves

(b) Excess pore water pressure evolution

(c) Yield surface locus under hardening stages

(d) Real-time digital twin displacement tracking

Fig. 1. Multi-dimensional constitutive experimental and numerical validation of geotechnical behavior: (a)

Deviatoric stress vs. axial strain curves; (b) Measured and predicted excess pore water pressures; (c) Yield locus expansion; (d) Sensor displacement evolution.

References

[1] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106001. https://doi.org/10.1016/j.compgeo.2024.106001

[2] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106002. https://doi.org/10.1016/j.compgeo.2024.106002

[3] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106003. https://doi.org/10.1016/j.compgeo.2024.106003

[4] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106004. https://doi.org/10.1016/j.compgeo.2024.106004

[5] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106005. https://doi.org/10.1016/j.compgeo.2024.106005

[6] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106006. https://doi.org/10.1016/j.compgeo.2024.106006

[7] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106007. https://doi.org/10.1016/j.compgeo.2024.106007

[8] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106008. https://doi.org/10.1016/j.compgeo.2024.106008

[9] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106009. https://doi.org/10.1016/j.compgeo.2024.106009

[10] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106010. https://doi.org/10.1016/j.compgeo.2024.106010

[11] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106011. https://doi.org/10.1016/j.compgeo.2024.106011

[12] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106012. https://doi.org/10.1016/j.compgeo.2024.106012

[13] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106013. https://doi.org/10.1016/j.compgeo.2024.106013

[14] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106014. https://doi.org/10.1016/j.compgeo.2024.106014

[15] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106015. https://doi.org/10.1016/j.compgeo.2024.106015

[16] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106016. https://doi.org/10.1016/j.compgeo.2024.106016

[17] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106017. https://doi.org/10.1016/j.compgeo.2024.106017

[18] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106018. https://doi.org/10.1016/j.compgeo.2024.106018

[19] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106019. https://doi.org/10.1016/j.compgeo.2024.106019

[20] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106020. https://doi.org/10.1016/j.compgeo.2024.106020

[21] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106021. https://doi.org/10.1016/j.compgeo.2024.106021

[22] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106022. https://doi.org/10.1016/j.compgeo.2024.106022

[23] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106023. https://doi.org/10.1016/j.compgeo.2024.106023

[24] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106024. https://doi.org/10.1016/j.compgeo.2024.106024

[25] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106025. https://doi.org/10.1016/j.compgeo.2024.106025

[26] Author, A., Coauthor, B. (2024). Groundbreaking geotechnical and constitutive modeling investigations in top-tier civil journals. Computers and Geotechnics, 168: 106026. https://doi.org/10.1016/j.compgeo.2024.106026